

Boss!

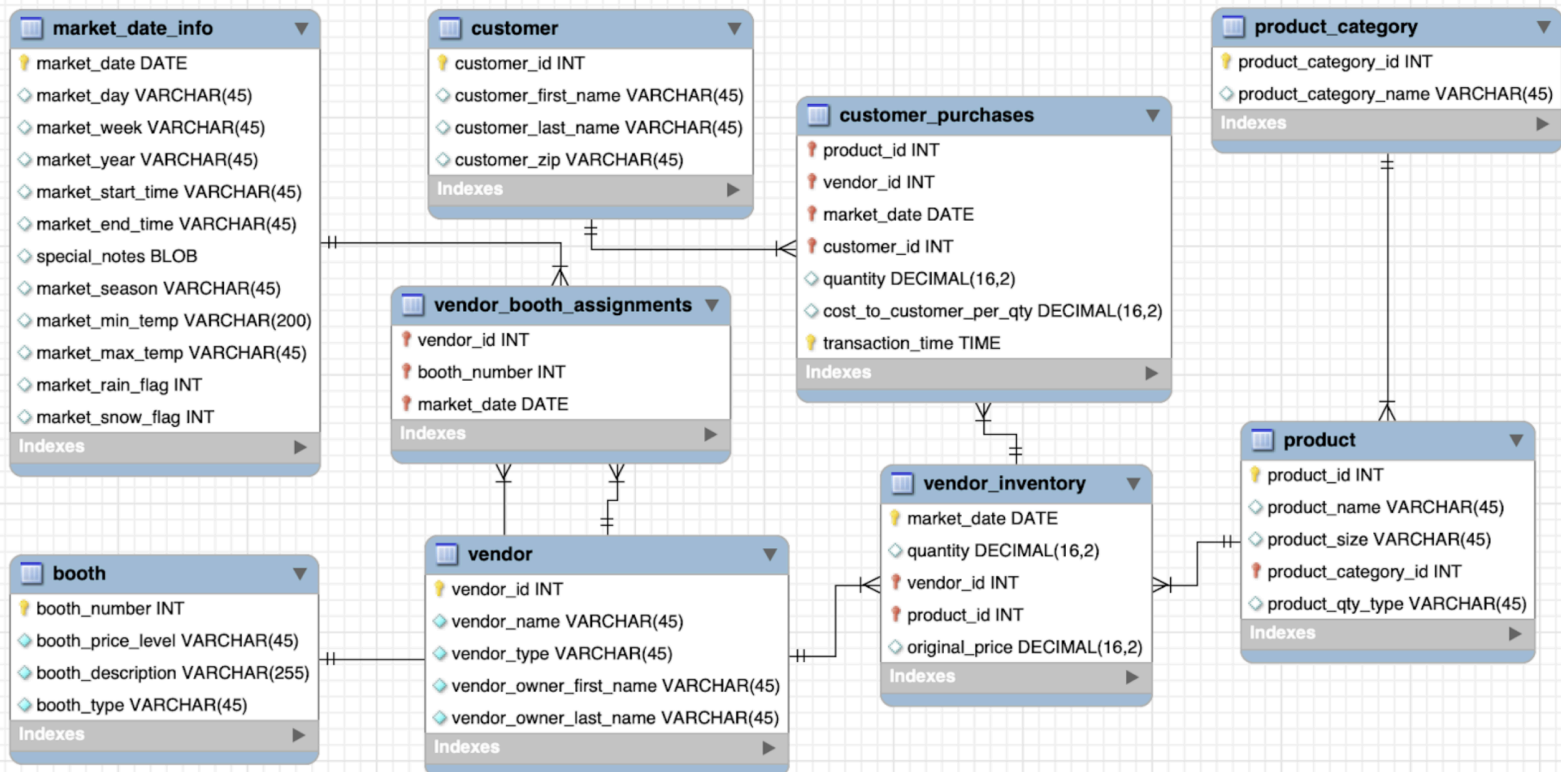
What can travel around the world
while staying in a corner?

A stamp

WhatsApp link

<https://chat.whatsapp.com/EvVhJ555AGW1cJaeKI0USe>

Dataset



Window Function

→ Aggregation performed at Group level but displayed at row level.

Salary normalization

e-id	e-name	D.O.B	salary	avg-sal	mid cycle Hike
1	Alpha	2020	60,000	72,500	15%.
2	Beta	2021	50,000	72,500	20%.
3	Gamma	2021	70,000	72,500	10%.
4	Sigma	2025	1,10,000	72,500	0%.

Amazon Great Indian Festival

Day	Rev. per-day	Cumm. Rev
1	250 M	250 M
2	200 M	450 M
3	100M	550 M

Window Function

Aggregated

- Sum
- Max
- Min
- Count
- Average

Analytical

- Row-number
- Rank
- dense rank
- n^{th} value
- n -tile
- lag / lead

Employees

ID	name	Salary	sum-sal
1	A	100	2100
2	B	200	2100
3	C	300	2100
4	D	400	2100
5	E	500	2100
6	F	600	2100

① using subquery

② using window function

↑ Higher Tc

↓ O(n)

Syntax vs Execution Order

Syntax Order	Execution Order
Select	From
From	where
where	group by
Group by	aggregated cols
Having	having
order by	Window clause
limit	Select
offset	order by
	offset
	limit

Over clause

over ()

↳ Partition By col (a)

↳ Order by col (b)

over ()

over (a) , over (b) , over (a, b)

Syntax

Select

fn_name () over ()

from table

where

over()

It will treat the entire table as one logical unit and apply the function for the logical unit

ID	name	Salary	Sum-sal
1	A	100	2100
2	B	200	2100
3	C	300	2100
4	D	400	2100
5	E	500	2100
6	F	600	2100

logical unit.

Query

Sum = 2100

Select *,

sum (salary) over()

from employee;

Partition By → Group By creates logic units based on column passed as parameter

id	name	Salary	Dept.	Designation	avg-sal
1	A	100	IT	Intern	133.33
2	B	200	IT	SDI	133.33
3	C	300	Admin	Manager	350
4	D	400	Admin	S. Manager	350
5	E	500	HR	S. Manager	500
6	F	100	IT	Intern	133.33

Query

Select *,

avg (salary) over (partition by dept) as avg-sal
from employees;

→ We still scan the table only once

→ We use $O(k)$ space complexity where k is your no. of logical partitions

Order By with Analytical → sorts the data based on ^{column}

↳ Row-number → logical unique no. for each row

↳ Rank

↳ dense rank

Id	name	Salary	Dept.	Designation	R.no	Rank	d-Rank
1	A	100	IT	Intern			
2	B	200	IT	SDI			
3	C	300	Admin	Manager			
4	D	400	Admin	S. Manager			
5	E	500	HR	S. Manager			
6	F	100	IT	Intern			

Query

select *,

row-number() over (order by salary desc)

as r-no

from employees;

id	name	Salary	Dept.	Designation	R.No	Rank	d-Rank
5	E	500	HR	S. Manager	1	1	1
4	D	400	Admin	S. Manager	2	2	2
3	C	300	Admin	Manager	3	3	3
2	B	200 300	IT	SDI	4	3	3
1	A	100	IT	Intern	5	5	4
6	F	100	IT	Intern	6	5	4

Rank skips values if it sees duplicates

However Rank dense does not skip value when you see any duplicates.

Window Frames

Use both partition by and order by together

→ unbounded preceding

→ current row

→ unbounded following

Visualize

id	name	Salary	Dept.	Designation
1	A	100	IT	Intern
2	B	200	IT	SDI
3	C	300	Admin	Manager
4	D	400	Admin	S. Manager
5	E	500	HR	S. Manager
6	F	100	IT	Intern

→ partition based
on department

id	name	Salary	Dept.	Designation
1	A	100	IT	Intern
2	B	200	IT	SDI
6	F	100	IT	Intern
3	C	300	Admin	Manager
4	D	400	Admin	S. Manager
5	E	500	HR	S. Manager

→ order by name.

determines my
window frame

Announcement

→ substitute teacher teaching.

only for next class.